

IMDB Movie Rating Analysis

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Project Objectives

Predict IMDb Scores

Using genre, director, actors, language & country

Feature Importance

Identify which attributes most influence ratings

Evaluate Models

MSE, MAE & R^2 across three regression algorithms

EDA & Visualisation

Uncover trends via histograms, bar charts & box plots



Dataset at a Glance



5,043 Movies

28 columns of metadata

Key Features

Director, actors, genre, language, country, budget, gross, duration

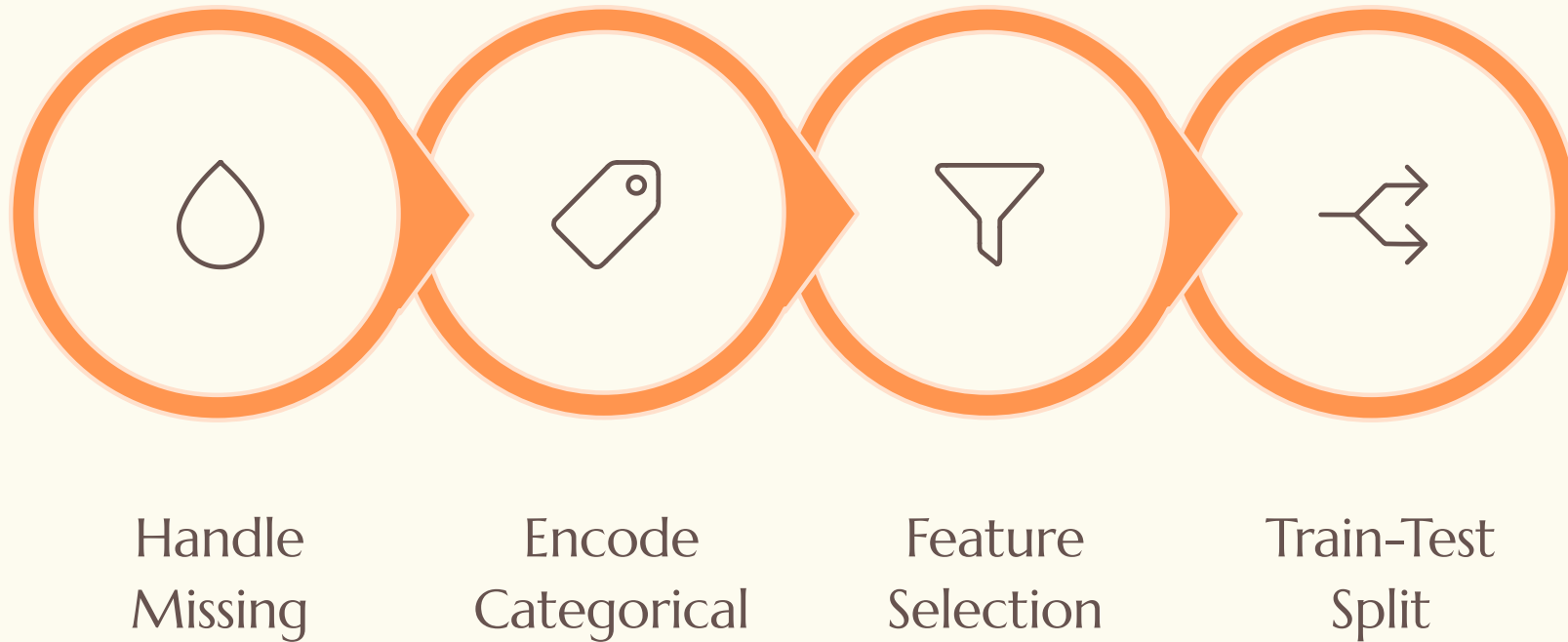
Target Variable

IMDb Score (0–10)

Source

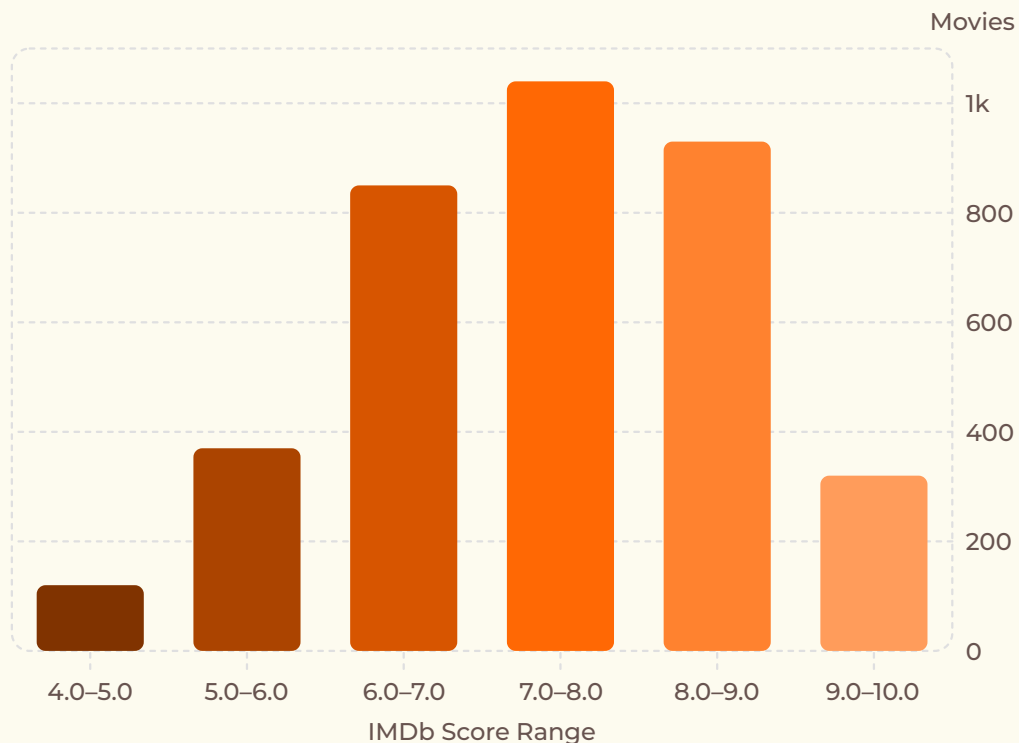
Kaggle — Movie Metadata CSV

Data Preprocessing Pipeline



Clean, encoded data ensures reliable model training and evaluation.

EDA: Score Distribution & Language



Key Insights

- Scores peak at **7.0-7.5** — slightly right-skewed
- **English** dominates language count by a wide margin
- Most movies score between **5 and 8**

EDA: Content Rating & Top Directors



Content Rating

PG-13 & R ratings dominate; **Approved** has the highest median IMDb score (7.5)

Top Directors by Movie Count

Steven Spielberg

25 films

Woody Allen / Clint
Eastwood

19 films each

Ridley Scott

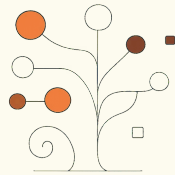
17 films

Machine Learning Models



Linear Regression

Baseline model. Assumes linear feature–score relationship. Simple & interpretable.



Decision Tree

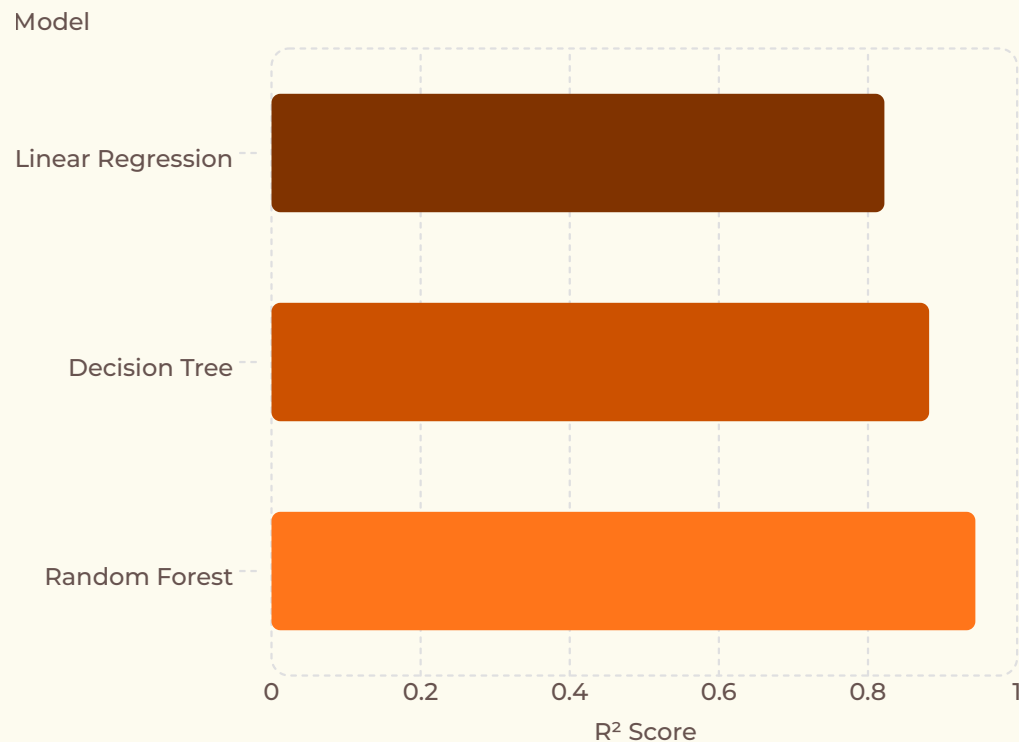
Captures non-linear patterns. Splits on budget, director rating, reviews.



Random Forest

Ensemble of 100 trees. Averages predictions to reduce overfitting.

Model Performance Comparison



Random Forest Wins

MSE: 0.382

Lowest error

MAE: 0.410

~0.41 point avg error

R²: 0.944

94.4% variance explained

Feature Importance — Random Forest



Conclusion & Future Scope

Conclusion

Random Forest achieved $R^2 = 0.944$ — best performer. Budget & director popularity are the strongest rating predictors.

Future Directions

- XGBoost / LightGBM
- Sentiment Analysis on plot text
- SHAP / LIME explainability
- Real-time Streamlit / Flask app